Lecture 10 (2026-07-12)

Quick Summary

The lecture covers Profitability Ratios, divided into two categories: Margin Ratios (Gross Profit, Operating Profit, EBIT, and PAT) and Return Ratios (ROE, ROCE). It also covers Liquidity Ratios (Current, Quick, and Absolute), and Turnover (Activity) Ratios. The segment also introduces Evaluation Ratios (P/E, P/B) and the Indirect Method of calculating Cash Flow from Operations, including specific calculations for Cash Flow from Investing and Cash Flow from Financing. It further distinguishes between Standalone and Consolidated statements and identifies "red flags" in cash flow analysis.

Key Definitions

- **Profitability Ratios**: Categorized into **Margin Ratios** (e.g., Gross Profit Margin, Operating Profit Margin, EBIT Margin, PAT Margin) and **Return Ratios** (e.g., ROE, ROCE).
- **Operating Profit**: Equivalent to **EBIT** (Earnings Before Interest and Tax). This represents firm-level profit.
- **Net Profit**: Profit available specifically to the shareholders (PAT).
- **NOPAT (Net Operating Profit After Tax)**: Operating profit minus tax. It does **not** include the deduction of interest.
- **Capital Employed**: The total long-term capital used by the company, calculated as **Equity + Debt**.

  Earnings Per Share (EPS):
  - **Basic**: Net Profit (after interest and tax, less preference dividends) divided by the number of equity shares.
  - **Diluted**: Includes all shares that can be converted into equity in the future (e.g., convertible bonds, warrants, options, ESOPs).
- **Current Ratio**: Ratio of Current Assets to Current Liabilities.
- **Quick Ratio**: A measure of liquidity excluding assets that are not easily converted to cash (Prepaid expenses and Inventory).
- **Absolute (Cash) Ratio**: A ratio of assets that can be converted to cash immediately (Cash + Marketable Securities) to Current Liabilities.
- **Turnover (Activity) Ratios**: Measure the efficiency of assets/liabilities, such as how many times an item (like Accounts Receivable) is "turned over" relative to sales.
- **Cash Conversion Cycle (CCC)**: A calculation of the time it takes to convert investments in inventory and other resources into cash from sales.
- **Price to Earnings (P/E) Ratio**: Measures the price paid for every unit of earning per share. It reflects how much investors are willing to pay based on growth expectations.
- **Price to Book Value (P/B) Ratio**: Measures the price paid for the "book value" of the assets (Assets minus Liabilities). Useful for asset-heavy companies or banking.
- **Cash Flow (Indirect Method)**: A method of calculating Cash Flow from Operations by starting with a profit figure (Operating Profit, EBIT, or PAT) and adjusting for non-cash items and non-operating items.
- **Debentures/Bonds**: Debt instruments traded on the stock market. They are interchangeable terms for debt with an interest charge attached.
- **Treasury Shares**: Shares bought back by the company and removed from the market; they are no longer owned by anyone and are not added to the total share count.
- **Standalone**: Financial statements prepared for a single entity without including the results of subsidiaries or mergers.
- **Consolidated**: Financial statements that include the results of the parent company and all its subsidiaries/merbers.
- **Impairment**: A reduction in the carrying amount of an asset (e.g., due to obsolescence or a drop in market value) that represents a loss in the value of the investment.
- **Government Grants**: Income received from the government; these are removed from operating profit as they are not considered "operating" income.

- **Exceptional Items**: Non-recurring items (e.g., fire loss, large one-off dividends, windfall gains/losses) that do not happen on a regular basis. These should be separated from regular operating figures.
- **Provision**: A non-cash item (e.g., Provision for bad debts or diminution) set aside for future expenses.

Worked Examples

- **Calculating Return on Equity (ROE)**:
- Formula: (Net Profit - Preference Dividends) / (Shareholders' Equity + Reserves and Surplus)
- **Reasoning**: Preference dividends must be subtracted from Net Profit because they are paid to preference shareholders before equity shareholders receive any distribution. Only the remainder is "available" to equity shareholders.
- **Calculating Return on Capital Employed (ROCE)**:
- Formula: Operating Profit / Average Capital Employed
- **Reasoning**: The denominator represents the total long-term capital used to fund the company.
- **Calculating Earnings Per Share (EPS)**:
- **Basic EPS**: (Net Profit - Preference Dividends) / Number of Equity Shares.
- **Diluted EPS**: (Net Profit - Preference Dividends) / (Equity Shares + Convertible Bonds + Warrants + Options + ESOPs + Partially Paid-up Shares).
- **Reasoning**: Diluted EPS is a more conservative measure because it accounts for the future possibility of all potential equity-convertible instruments becoming shares.
- **Liquidity Ratio Calculations**:
- **Current Ratio**: Current Assets / Current Liabilities.
- **Quick Ratio**: (Current Assets - Prepaid Expenses - Inventory) / Current Liabilities. (Note: Accounts Receivable may also be excluded depending on the level of control the firm has over those debts).
- **Absolute Ratio**: (Cash in hand + Bank + Marketable Securities) / Current Liabilities.
- **Turnover Ratio Calculations**:
- **Accounts Receivable Turnover**: Credit Sales / Average Accounts Receivable.
- **Accounts Payable Turnover**: Credit Purchases (or Trade Creditors) / Average Accounts Payable.
- **Raw Materials Turnover**: Cost of Goods Sold (COGS) / Raw Materials.
- **Finished Goods Turnover**: Sales / Finished Goods.
- **Calculating Cash Conversion Cycle (CCC)**:
- **Step 1**: Calculate Accounts Receivable Days: $365 / \text{Accounts Receivable Turnover}$.
- **Step 2**: Calculate Inventory Days: $365 / \text{Inventory Turnover}$.
- **Step 3**: Calculate Accounts Payable Days: $365 / \text{Accounts Payable Turnover}$.
- **Step 4**: Calculate CCC: $(\text{Accounts Receivable Days} + \text{Inventory Days}) - \text{Accounts Payable Days}$.
- **Converting Operating Profit to Cash Flow from Operations (Indirect Method)**:
- **Step 1**: Start with a profit figure (Operating Profit, EBIT, or PAT).
- **Step 2**: Subtract "Other Income" (e.g., from sale of land/junk) to isolate operating-only items.

  Step 3: Add back "Non-Cash" expenses:
  - **Depreciation**: Fall in value of fixed assets (no cash involved).
  - **Amortization**: Fall or fall in value of intangible assets (no cash involved).
  - **Bad Debts**: No cash involved in the realization of these costs.
  - **Provisions**: Created for future expenses where cash hasn't left yet.
  - **Impairment**: Included in the add-back as it represents a non-cash drop in asset value.

  Step 4: Subtract **Taxes** (considered operating in nature) and ignore **Interest** (considered financing in nature) for the purposes of determining the transition to the financing section.
  - **Exception (Starting from EBT)**: If starting from EBT (Earnings Before Tax), interest is already deducted as a negative number in the P&L. Therefore, to reach the correct operating figure, interest must be **added** back.
  - **Non-Operating Income**: Items like **Interest Income** or **Dividend Income** must be **subtracted** as they are not operating items.

Step 5: Adjust for Working Capital:

  - **Current Assets (CA)**: If there is an **increase** in CA (excluding cash, e.g., Inventory/Receivables), **subtract** from the profit (cash is "locked"). If there is a **decrease**, **add** to the profit.
  - **Current Liabilities (CL)**: If there is an **increase** in CL (e.g., Accounts Payable), **add** to the profit (cash is "saved"). If there is a **decrease**, **subtract**.

- **Categorizing Specific Items in Cash Flow Statements**:
- **Income Tax**: Categorized as **Operating**.
- **Mining Rights**: Categorized as **Investment** (Investing activity).
- **Dividends Paid**: Categorized as **Financing**.
- **Dividends Received**: Categorized as **Investment** (considered investment income).
- **Interest Expense**: Categorized as **Financing**.

Non-Cash Items (Depreciation, Amortization, Bad Debts, Impairment, Provisions): Added back to Operating Profit.

  - **Reasoning**: These are non-cash items. They were subtracted from the profit to reach the P&L total, but since no cash left the company, they must be "added back" to reveal the actual cash position of the operations.

- **Calculating Cash Flow from Investing (CFI)**:
- **Formula**: $(\text{Other Income} + \text{Sales of Assets/Investments}) - (\text{Purchase of Assets} + \text{Investments} + \text{Acquisitions})$.
- **Components**: Purchase of assets includes tangible assets, intangible assets, and the acquisition of other companies (e.g., mining rights).
- **Note**: Investment is typically a negative activity because the "Purchase of Assets" is usually the largest figure.
- **Calculating Cash Flow from Financing (CFF)**:
- **Formula**: $(\text{Issue of Shares} + \text{Issue of Bonds/Debentures} + \text{Loans from Bank}) - (\text{Dividends} + \text{Interest} + \text{Buyback of Shares} + \text{Redemption of Loans/Preferences})$.
- **Calculating Total Cash Flow**:
- **Calculation**: $\text{Opening Cash} + \text{Cash Flow from Operations} + \text{Cash Flow from Investing} + \text{Cash Flow from Financing} = \text{Closing Cash}$.

Formulas & Rules

- **Profitability Margin Differentiation**:
- **Gross Profit Margin**: Reflects the cost of goods sold (COGS).
- **Operating Profit Margin**: Reflects the impact of operating expenses (e.g., salaries, overheads).
- **Margin Gap Analysis**:
- If Operating Profit Margin is high but PAT Margin is low, the discrepancy is caused by either **high interest payments** or a **high tax rate**.
- **Liquidity Ratio Logic**:
- **Current Ratio**: Target is typically 2:1, but this varies by industry.
- **Quick Ratio**: Excludes items that "don't move easily" (Prepaid Expenses and Inventory).
- **Absolute Ratio**: Only includes items that can be converted to cash immediately (Cash + Marketable Securities).
- **Turnover Calculation Rules**:

Numerator Selection:

  - Use **Credit Sales** for Accounts Receivable because AR only represents money not yet collected.
  - Use **Cost of Goods Sold (COGS)** for Raw Materials because they are valued at cost.
  - Use **Sales** for Finished Goods because they are ready to be sold at selling price; the cost is no longer the primary concern for the manager.

- **Denominator Selection**: Use **Average** figures (e.g., Average Accounts Receivable) to create a continuous measurement across the period, rather than a single point-in-time figure.
- **Cash Flow Adjustment Rules (Indirect Method)**:

- **Non-Cash Items**: Items like Depreciation, Amortization, Bad Debts, and Provisions must be **added back** to the profit figure because they were subtracted to reach the profit total but did not involve an actual outflow of cash.
- **Non-Operating Items**: Items like "Other Income" (from non-operating sources) must be **subtracted** from the operating profit to ensure the cash flow calculation reflects only the "operating" component.
- **Interest vs. Tax**: Interest is a **financing** expense; Tax is an **operating** expense.

  Working Capital Adjustments:
  - **Current Assets**: An increase is **subtracted** (cash is "locked"); a decrease is **added**.
  - **Current Liabilities**: An increase is **added** (cash is "saved"); a decrease is **subtracted**.
- **Cash Flow from Investing (CFI) Rules**:
- **Formula**: $(\text{Other Income} + \text{Sales of Assets}) - (\text{Purchase of Assets} + \text{Investment} + \text{Acquisition})$.
- **Acquisition Scope**: Includes tangible assets, intangible assets, and mining rights.
- **Cash Flow from Financing (CFF) Rules**:
- **Additions**: Issues of shares, issuance of bonds/debentures, and loans from banks.
- **Subtractions**: Dividends, interest payments, buybacks of shares, and redemption of loans/preferences.

Tricky Logic & Traps

- **Operating Profit vs. NOPAT**: Students often confuse these two. NOPAT is simply Operating Profit with tax removed; it does **not** subtract interest. In many reports, NOPAP is used as a substitute for operating profit, but the distinction regarding interest is critical.
- **Treatment of "Other Income"**: Technically, "Other Income" is not a result of operations. However, because it is usually a very small portion of total income, many firms include it in their operating profit figures.

  Identifying the Source of Profit Drop:
  - If the **Gross Profit Margin** falls, the issue is in the **COGS** (e.g., rising raw material costs like crude oil or rubber).
  - If the **Operating Profit Margin** falls, the issue is in **Operating Expenses** (e.g., rising salaries or increased benefits like Provident Fund).
- **Reserves and Surplus**: These are considered part of the "Owner's Equity" because they are profits not taken home by owners but retained/allocated for different purposes. They must be included in the denominator for equity-based calculations.
- **"Other" Factors in Profit**: A "Chaiwala" (small street vendor) may have a higher net profit margin than a large corporation because they do not pay taxes or significant rent, illustrating how tax and fixed costs impact the "Net" while not impacting the "Operating" margin.
- **Diluted vs. Basic EPS**: Use **Diluted** for a more conservative measure. Investors are concerned with the future, where convertible bonds, warrants, and ESOPs are likely to be exercised.
- **Liquidity Importance**: While the **Current Ratio** is the most common in exams, the **Absolute Ratio** is the most useful for a manager because it tracks the most immediate source of liquidity (cash).
- **Exclusion of Inventory/Prepaid in Quick Ratio**: These are excluded because they are not "moving" assets or cannot be converted to cash at will. Accounts Receivable are sometimes excluded if the company lacks control over the collection timeline.

  Raw Materials vs. Finished Goods Calculation:
  - Use **COGS** for Raw Materials because they are valued at cost.
  - Use **Sales** for Finished Goods because they are ready to be sold at selling price.

  The "Average" Calculation Dilemma: When only closing data is provided:
  - **Mathematical approach**: (Opening + Closing) / 2.
  - **Management/Decision-making approach**: If opening data is missing, assume Closing = Average.
  - **Logic**: If you assume the opening balance was 0, the calculation might be mathematically "correct" but leads to a poor interpretation of the data. Using the "ballpark figure" of the current year's data allows the manager to make better decisions even if it isn't perfectly precise.
- **Value Investing & P/B Ratio**: A low Price to Book Value ratio is only considered a "buy" signal if the company is fundamentally sound. A low P/B could be a trap if the company has fundamental issues or legal problems (e.g.,

"rinsed" book values or refund/fraud).

- **High P/E Ratio Interpretation**: A high P/E ratio does not automatically mean a company is "bad." It often indicates high **growth expectations** from the market.
- **"Locked" Cash in Cash Flow**: When calculating Cash Flow from Operations, an increase in current assets (other than cash, like inventory or receivables) must be **subtracted**. This is because that cash is "locked" in those assets and is not currently "flowing" into the bank.
- **Interest vs. Tax in Cash Flow Calculation**: When moving from the P&L to the Cash Flow statement, Interest is treated as a **Financing** item, while Tax is treated as an **Operating** item.
- **Working Capital & Cash Flow Dynamics**: An increase in Current Assets (like Inventory) "locks" cash in the operations, while an increase in Current Liabilities (like Accounts Payable) "saves" cash because the firm is not yet paying those obligations.
- **CFI as a Growth Indicator**: A negative value for Cash Flow from Investing (CFI) is often a sign of a growing company, as it means the expenditure on assets (purchases/acquisitions) outweighs the proceeds from selling them.
- **Debentures vs. Shares**: Debentures and Bonds are debt instruments; they pay interest. Shares are equity and pay dividends. Note: "Debenture" and "Bond" are often used interchangeably for debt traded on the market.
- **Dividend Nature**: Dividends are NOT an expense. They do not appear on the Statement of Profit and Loss (P&L) because they are a distribution of profit to owners, not a cost of doing business.
- **Treasury Shares**: When shares are bought back, they are removed from the market and become "Treasury Shares." They are not owned by anyone and are not included in the total share count.
- **The "Red Flag" of Cash Flow vs. Dividends**: A company that reports a negative Cash Flow from Operations (CFO) but is paying out large dividends is considered a **bad sign**. This indicates the company is paying out more than it is generating from its core operations.
- **EBT vs. EBIT Calculation Logic**: When using the Indirect Method, the treatment of interest depends on the starting point. If starting from **EBIT**, no action is needed for interest. If starting from **EBR** (Earnings Before Revenue/Tax), interest must be **added back** because interest is recorded as a negative figure in the P&L; adding it back reverses the deduction to show the operating figure.
- **Standardized Calculation of "Add-backs"**: Items like Depreciation, Amortization, and Bad Debts are always added back because they are non-cash expenses. **Impairment** is also added back as it is a non-cash loss in value.
- **Exclusion of Non-Operating items**: Items like **Interest Income** or **Dividend Income** should be subtracted during the calculation of Cash Flow from Operations because they are non-operating items.
- **Growth vs. Stagnation in Operations**: A positive and increasing Cash Flow from Operation (CFO) is generally a sign of a healthy, growing company.
- **The "No Cash" Trap**: A common trick question involves the conversion of **Debentures to Equity**. This transaction should **not** be reported on the Cash Flow statement because no actual cash is exchanged.
- **Difference between Dividend Received and Paid**: **Dividend Received** is categorized as an **Investing** activity (investment income), whereas **Dividend Paid** is categorized as a **Financing** activity.
- **Profit vs. Cash Flow Gap**: A company can show a net loss (negative profit) but still report a **positive Cash Flow from Operations**. This typically happens when there is a significant positive movement in **Working Capital** (e.g., inventory reduction or faster collection of receivables).
- **Exceptional Items Logic**: These are non-recurring items (one-off events like fire damage or windfall gains). Because they are not recurring, they should be identified and potentially removed from standard operating metrics to avoid skewing analysis.
- **Rule of Thumb for Non-Cash Items**: Any item that is non-cash (Depreciation, Amortization, Bad Debts, Impairment, Provisions) must be **added back** during the transition from profit to cash flow to "remove" the non-cash component.

Exam Pointers

- **Focus on Movement**: In the Cash Flow Statement, the "movement" of cash is a core concept. Ensure you can track how transactions in the P&L (like Depreciation or Interest) translate into the "Flow" in the Cash Flow Statement.
- **Indirect Method Impact**: Understand that the Indirect Method isn't just a list of adjustments; it is a way to strip away non-cash and non-operating items to see the pure "impact" of operations on the cash balance.

Sources

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